

Homework 3

PENG Qiheng

Student ID 225040065

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Problem 1.

(a) Prove (using words) whether the proposition

$$((a \vee b) \wedge (\neg a \vee c)) \rightarrow \neg(b \vee c)$$

is valid.

Solution. This proposition is **not** valid (not a tautology). Give a counterexample:

$$a = \text{F}, \quad b = \text{T}, \quad c = \text{F}.$$

Then

$$(a \vee b) = \text{T}, \quad (\neg a \vee c) = \text{T},$$

so the antecedent is true. But

$$\neg(b \vee c) = \neg(\text{T} \vee \text{F}) = \neg\text{T} = \text{F}.$$

Therefore “true $\rightarrow$ false” is false. So the proposition is not valid.

(b) Use propositional resolution to show that the clause set

$$\{p, q, \neg r, s\}, \quad \{\neg p, r, s\}, \quad \{\neg q, \neg r\}, \quad \{p, \neg s\}, \quad \{\neg p, \neg r\}, \quad \{r\}$$

is unsatisfiable.

Solution. Let the clauses be

$$C_1 = (p \vee q \vee \neg r \vee s),$$

$$C_2 = (\neg p \vee r \vee s),$$

$$C_3 = (\neg q \vee \neg r),$$

$$C_4 = (p \vee \neg s),$$

$$C_5 = (\neg p \vee \neg r),$$

$$C_6 = (r).$$

Resolve step by step:

$$C_3, C_6 \vdash (\neg q) \quad (R_1)$$

$$C_5, C_6 \vdash (\neg p) \quad (R_2)$$

$$C_4, R_2 \vdash (\neg s) \quad (R_3)$$

$$C_1, C_6 \vdash (p \vee q \vee s) \quad (R_4)$$

$$R_4, R_1 \vdash (p \vee s) \quad (R_5)$$

$$R_5, R_2 \vdash (s) \quad (R_6)$$

$$R_6, R_3 \vdash \square$$

We derived the empty clause $\square$, so the clause set is unsatisfiable.

Problem 2.

(a) How many of the disjunctions

$$p \vee \neg q, \quad \neg p \vee q, \quad q \vee r, \quad q \vee \neg r, \quad \neg q \vee \neg r$$

can be made simultaneously true?

Solution. All **five** can be true simultaneously. One satisfying assignment is

$$p = \text{T}, \quad q = \text{T}, \quad r = \text{F}.$$

Check:

$$p \vee \neg q = \text{T} \vee \text{F} = \text{T},$$

$$\neg p \vee q = \text{F} \vee \text{T} = \text{T},$$

$$q \vee r = \text{T} \vee \text{F} = \text{T},$$

$$q \vee \neg r = \text{T} \vee \text{T} = \text{T},$$

$$\neg q \vee \neg r = \text{F} \vee \text{T} = \text{T}.$$

So the maximum number is 5.

(b) In standard mathematics, can

$$x = y, \quad y = z, \quad z = -x$$

hold simultaneously? In propositional logic, consider

$$A \Rightarrow B, \quad B \Rightarrow C, \quad C \Rightarrow \neg A.$$

Decide:

- i. They cannot be true simultaneously in all cases?
- ii. They can be true simultaneously in some cases?
- iii. They can be true simultaneously in all cases?

Solution. **Mathematics part.** From $x = y$ and $y = z$, we get $x = z$. Together with $z = -x$, we get $x = -x$, hence $2x = 0$, so $x = 0$. Then $y = z = 0$. Therefore they **can** hold simultaneously, but only for

$$x = y = z = 0.$$

Propositional logic part. Let

$$F = (A \Rightarrow B) \wedge (B \Rightarrow C) \wedge (C \Rightarrow \neg A).$$

Truth table:

A	B	C	$A \Rightarrow B$	$B \Rightarrow C$	$C \Rightarrow \neg A$	F
T	T	T	T	T	F	F
T	T	F	T	F	T	F
T	F	T	F	T	F	F
T	F	F	F	T	T	F
F	T	T	T	T	T	T
F	T	F	T	F	T	F
F	F	T	T	T	T	T
F	F	F	T	T	T	T

Conclusions:

- i. Yes. They are **not** true in all cases (for example $A = B = C = T$ makes F false).
- ii. Yes. They are true in some cases (for example $A = B = C = F$ makes F true).
- iii. No. They are not true in all cases.